

ABERDARE TECHNICAL COLLEGE

ICT LEVEL 5

MODULE 3

PRACTICAL 1

UNIT: Computer Networking

UNIT CODE: 0612 451 07A

ELEMENT 1:DESIGN COMPUTER NETWORK

Duration: 3 HOURS

Instructions to Candidates:

- ✓ *Save your Packet Tracer file as YourName_PracticalExam.pkt before you begin.*
- ✓ *Complete all tasks in the order given. Do not delete devices or configurations from earlier tasks — the examiner will review your final file.*
- ✓ *Take a screenshot after each successful verification step and paste it into a Word document named YourName_Screenshots.docx, labelled by task number.*
- ✓ *Complete the written tasks directly in the same Word document.*
- ✓ *You have been provided with the following material*
 - **Desktop computer installed with packet tracer**
 - candidate tool**
- ✓ *Ensure you check that you have steady power supply and computer is working*

SECTION A: Star Topology

Task A1 – Build the Topology

1. Place 1 switch (2960) and 3 PCs in the workspace, arranged visually as a star.
2. Connect each PC to the switch using copper straight-through cables, using ports FastEthernet0/1, 0/2, and 0/3.
3. Confirm all link lights are green (link up) in Simulation or Realtime mode.

Task A2 – IP Configuration

Configure static IP addresses on the three PCs:

- PC1: 192.168.1.10 / 255.255.255.0
- PC2: 192.168.1.11 / 255.255.255.0
- PC3: 192.168.1.12 / 255.255.255.0

Task A3 – Verification

1. From PC1's Command Prompt, run ping 192.168.1.11 and ping 192.168.1.12.
2. Confirm both pings show successful replies (0% loss).
3. Attach a screenshot of the successful ping output.

Task A4 – Discussion Task

1. Write, in your own words, an explanation of why all three PCs must be in the same subnet in order to ping each other directly, without a router.
2. Describe what happens at Layer 2 and Layer 3 when this condition is not met.

SECTION B: Connecting Two Networks with a Router

Task B1 – Build the Topology

1. Place 2 switches, each connected to 2 PCs, forming two separate LANs (LAN A and LAN B).
2. Place 1 router 2911 between the two switches.

3. Connect the router's GigabitEthernet0/0 to Switch A and GigabitEthernet0/1 to Switch B, using appropriate cables.
4. Confirm all links show green/up status.

Task B2 – IP Addressing Scheme

Configure the following:

- LAN A, PC1: 192.168.1.10 / 255.255.255.0, gateway 192.168.1.1
- LAN A, PC2: 192.168.1.11 / 255.255.255.0, gateway 192.168.1.1
- LAN B, PC1: 192.168.2.10 / 255.255.255.0, gateway 192.168.2.1
- LAN B, PC2: 192.168.2.11 / 255.255.255.0, gateway 192.168.2.1

Task B3 – Router Configuration via CLI

Using the router's CLI tab, configure both interfaces with IP addresses and enable them; use the configuration commands.

```
Router(config)# interface GigabitEthernet0/0
```

```
Router(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
Router(config-if)# no shutdown
```

```
Router(config)# interface GigabitEthernet0/1
```

```
Router(config-if)# ip address 192.168.2.1 255.255.255.0
```

```
Router(config-if)# no shutdown
```

Task B4 – Verification

1. Run show ip interface brief on the router and confirm both interfaces show up/up.
2. From a PC in LAN A, ping a PC in LAN B (e.g., ping 192.168.2.10).
3. Attach a screenshot of both the show ip interface brief output and the successful cross-LAN ping.

Task B5 – Troubleshooting Task

The examiner will disable one router interface using shutdown before you begin this task, without telling you which one.

1. Identify the fault using show ip interface brief and correct it.
2. Write a brief description of how you identified and fixed the fault.